



南开大学 作业纸

系别 计算机学院 班级 2-1 姓名 陆皓皓 22110444 第 1 页

1. 设长方体在第一卦限的顶点为 (x, y, z)

$$V = 8xyz$$

$$\text{限制: } \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$L = 8xyz + \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \right)$$

$$\begin{cases} F'_x = 8yz + 2\lambda \frac{x}{a^2} = 0 \\ F'_y = 8xz + 2\lambda \frac{y}{b^2} = 0 \\ F'_z = 8xy + 2\lambda \frac{z}{c^2} = 0 \\ \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \end{cases}$$

$$\Rightarrow \frac{a}{\sqrt{3}} x = \frac{a}{\sqrt{3}}, y = \frac{b}{\sqrt{3}}, z = \frac{c}{\sqrt{3}}$$

$$\therefore V = 8xyz = \frac{8abc}{3\sqrt{3}} = \frac{8}{9}\sqrt{3}abc$$

2. 1° $L = x^2 + y^2 + z^2 + \lambda(2 - x^2 - y^2) + \mu(x + y + z - 1)$

(此时求导为 $\sqrt{x^2 + y^2 + z^2}$)

$$\begin{cases} L'_x = 2x - 2\lambda x + \mu = 0 \\ L'_y = 2y - 2\lambda y + \mu = 0 \\ L'_z = 2z + \lambda + \mu = 0 \\ x^2 + y^2 + z^2 = 2 \\ x + y + z = 1 \end{cases}$$

$$\Rightarrow \lambda = 1 \text{ 或 } x = y$$

$$\text{若 } \lambda = 1 \Rightarrow x = 0, z = -\frac{1}{2}, \text{ 舍去}$$

$$\text{若 } x = y \Rightarrow x = y = \frac{\pm\sqrt{3}-1}{2}, z = 2 \mp \sqrt{3}$$

$$\therefore M_1(\frac{\sqrt{3}-1}{2}, \frac{\sqrt{3}-1}{2}, 2-\sqrt{3}), M_2(\frac{-\sqrt{3}-1}{2}, \frac{-\sqrt{3}-1}{2}, \sqrt{3}+2)$$

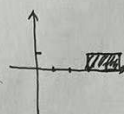
$$L_{\max} = \sqrt{9+5\sqrt{3}}, L_{\min} = \sqrt{9-5\sqrt{3}}$$

3. (1) A

$$0 \leq x+y \leq 1 \Rightarrow (x+y)^3 \leq (x+y)^2$$

$$\therefore I_1 \geq I_2, \text{ 选 A}$$

(2) B $3 \leq x \leq 5, 0 \leq y \leq 1$



$$\ln(x+y) \leq [\ln(x+y)]^2$$

$$x+y \in [3, 6] \quad \ln(x+y) > 1$$

$$\therefore [\ln(x+y)]^2 > \ln(x+y)$$

$$\therefore I_1 < I_2, \text{ 选 B}$$

$$(3) B \quad I = \lim_{r \rightarrow 0} \frac{1}{\pi r^2} \iint_{x^2+y^2 \leq r^2} e^{x^2-2y^2} \cos(\pi xy) d\sigma, I = c$$

$$I = \lim_{r \rightarrow 0} \frac{1}{\pi r^2} \iint_{x^2+y^2 \leq r^2} e^{x^2-2y^2} \cos(\pi xy) d\sigma$$

$$= \lim_{r \rightarrow 0} \frac{1}{\pi r^2} \int_0^{2\pi} \int_0^r e^{\rho^2 \cos^2 \theta - 2\rho^2 \sin^2 \theta} \cos(\pi \rho^2 \cos \theta \sin \theta) \rho d\rho d\theta$$

$$= \lim_{r \rightarrow 0} \int_0^{2\pi} \cos(\pi \rho^2 \cos \theta \sin \theta) d\theta$$

$$r \rightarrow 0, x^2 - 2y^2 \rightarrow 0, \pi xy \rightarrow 0$$

$$\therefore I = e^0 \cdot 1 = 1 \text{ 选 B}$$

(4) B $\iint_D xy \, dx \, dy$, $0 \leq y \leq x^2$, $0 \leq x \leq 1$

$$\int_0^1 xy \, dx \, dy = \int_0^1 x \, dx \int_0^{x^2} y \, dy$$

$$= \int_0^1 \left(x \cdot \frac{1}{2} y^2 \Big|_0^{x^2} \right) dx$$

$$= \int_0^1 x \cdot \frac{1}{2} x^4 \, dx$$

$$= \int_0^1 \frac{1}{2} x^5 \, dx$$

$$= \frac{1}{12} x^6 \Big|_0^1 = \frac{1}{12}$$

选 B

(5) A $0 \leq y \leq x^2$, $|x| \leq 2$

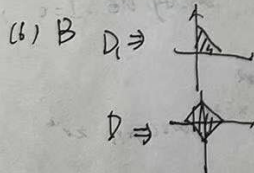
$$\iint_D xy^2 \, dx \, dy$$

$$= \int_{-2}^2 x \, dx \int_0^{x^2} y^2 \, dy$$

$$= \int_{-2}^2 x \, dx \left(\frac{1}{3} y^3 \Big|_0^{x^2} \right)$$

$$= \int_{-2}^2 x \cdot \frac{1}{3} x^6 \, dx$$

$$= \frac{1}{24} x^8 \Big|_{-2}^2 = \frac{1}{24} (2^8 - (-2)^8) = 0. \text{ 选 A}$$



$$\therefore \frac{D_1}{D} = \frac{1}{4} \text{ 选 B}$$

选 B